



Foundations of Business Analytics (for GenAI Managers)

Session 1: From Business Questions to Metrics

The Turing Test (Can Machine think? A. M. Turing, 1950)

- Requires
 - Natural language
 - Knowledge representation
 - Automated reasoning
 - Machine learning
 - (vision, robotics) for full test

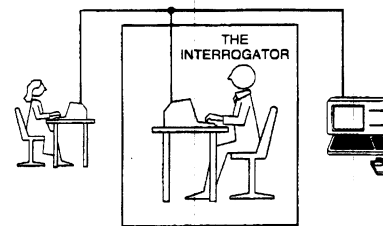
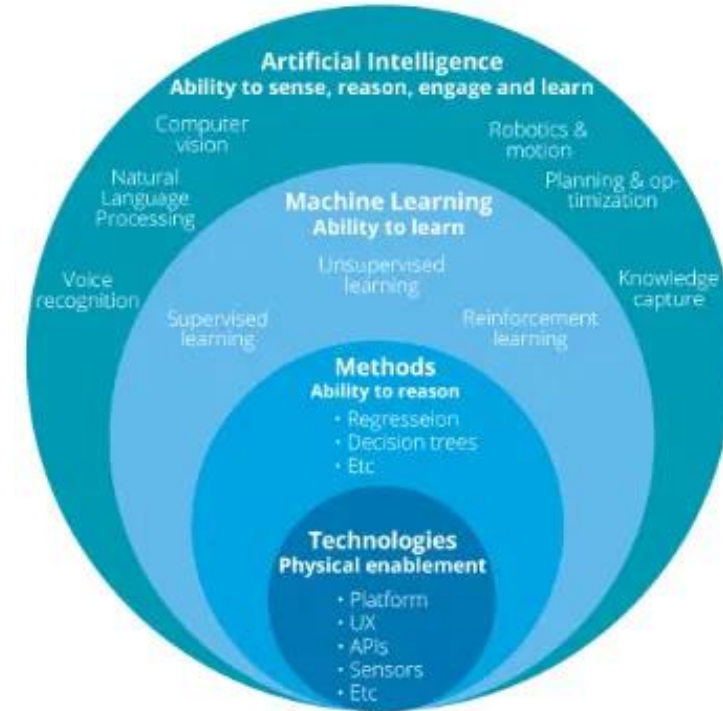


Figure 1.1 The Turing test.



AI Overview

Source: Deloitte



Artificial Intelligence (AI)

- Artificial narrow intelligence (ANI)
 - ANI is AI that specializes in one area. There is AI that can beat the world chess champion in chess, but that is the only thing it does.
- Artificial general intelligence (AGI)
 - AGI refers to a computer that is as smart as a human across the board—a machine that can perform any intellectual task that a human being can.
- Artificial superintelligence (ASI)
 - Oxford philosopher and leading AI thinker Nick Bostrom defines superintelligence as “an intellect that is much smarter than the best human brains in practically every field, including scientific creativity, general wisdom, and social skills.”

AI in the Workplace

Recent research shows that even non-routine tasks can be performed by algorithms (ML + Big Data)

Resume screening (Cowgill 2018)



Scanning CT images (Arbabshirani et al. 2018)

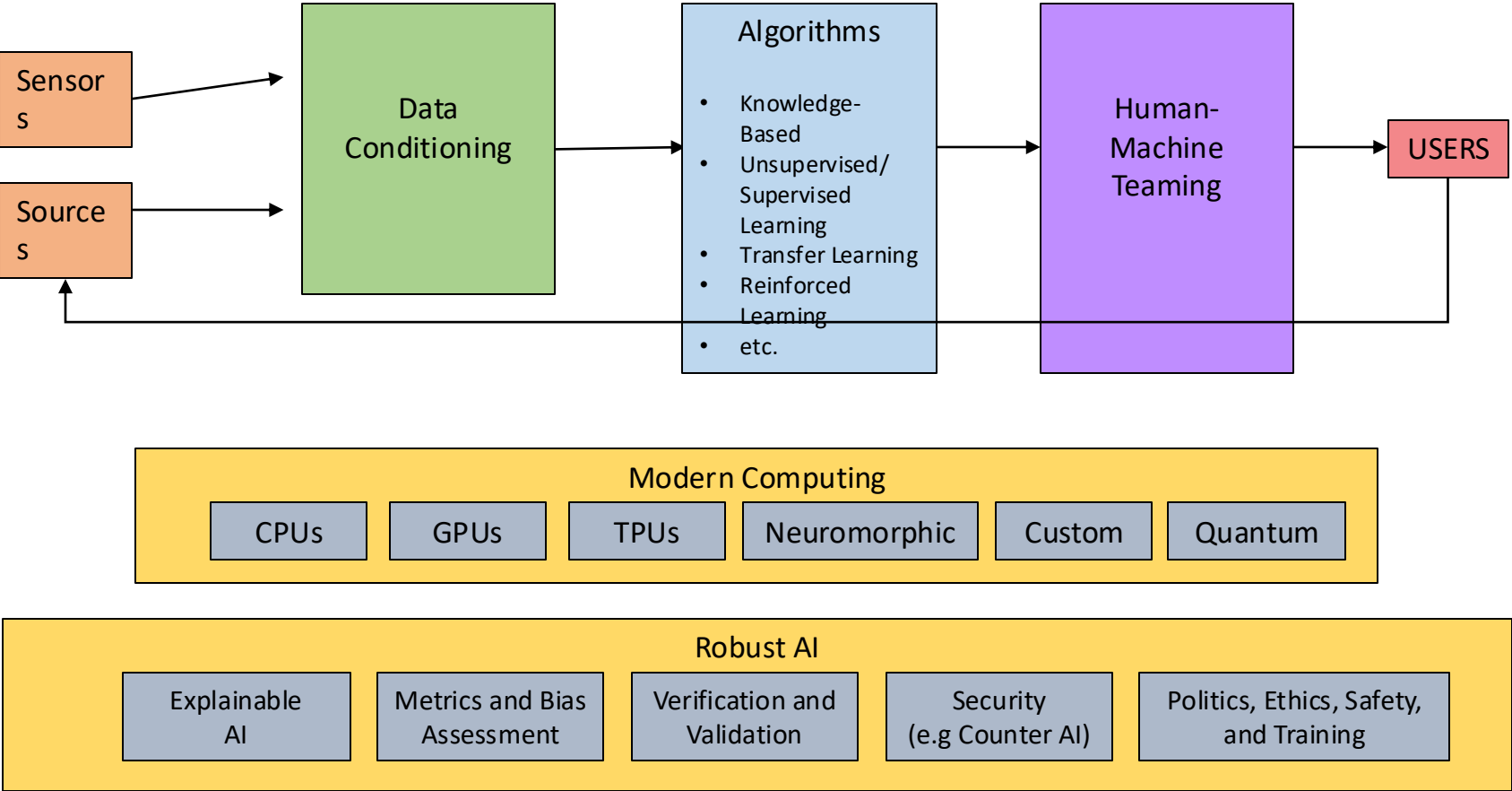


- Review judicial trials (



- 47% of US employment at risk due to automation (Frey and Osborne 2013)

High Level AI Architecture



Source: MIT Lincoln Laboratory

Today's Objectives

1

See the analytics workflow end-to-end

2

Build a KPI tree for a real decision

3

Draft a mini data dictionary

4

Use GenAI as an assistant—verify everything

Why Analytics Now (in a GenAI world)?

- Data volume & velocity up; experimentation is easier than ever
- GenAI can draft, summarise, and explore—but cannot replace judgment
- What hasn't changed: incentives, trade-offs, customers, and accountability

Big Data

- Collection of large datasets that cannot be processed using traditional techniques like regressions.
- Refers to both structured and unstructured data.
- Holds enormous promise to improve decision-making in organizations.

Characteristics of Big Data



1 Data is growing faster than ever before. By the year 2020,
about 1.7 megabytes
of new information will be created every second for every human being on the planet.

2 By 2025,
180 Zettabytes
data will be produced annually.
(That's 180 + 21 zeros!)



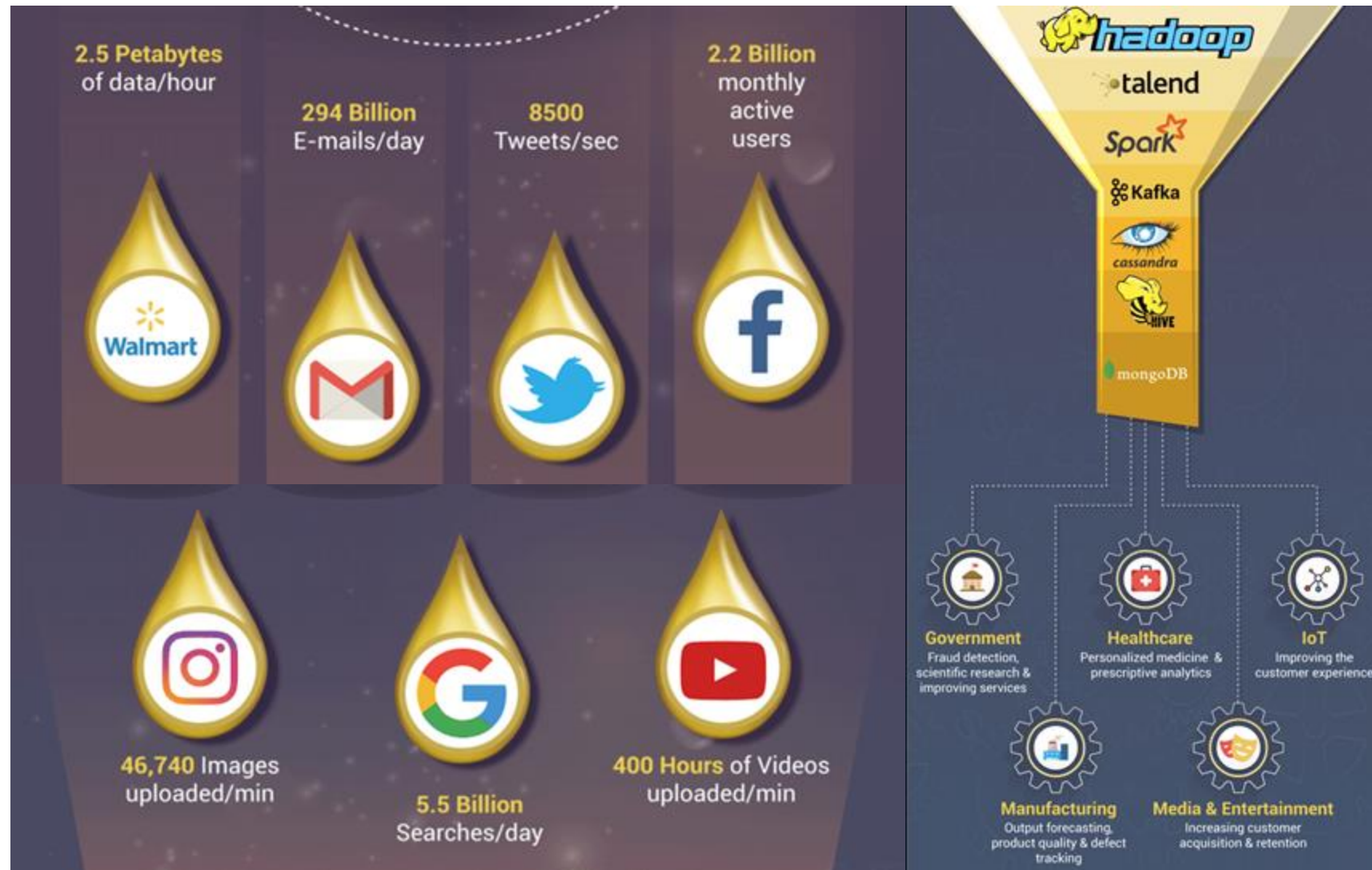
3 The data gathered isn't just text, but videos and photos, too.
On YouTube alone,
300 hours
of video are uploaded every minute.

4 In just 5 years, the number of connected devices in the world will be more than
50 billion—
all of which will create data that can be shared, collected and analyzed.



5 By 2020, at least
a third of all data
will pass through the cloud.







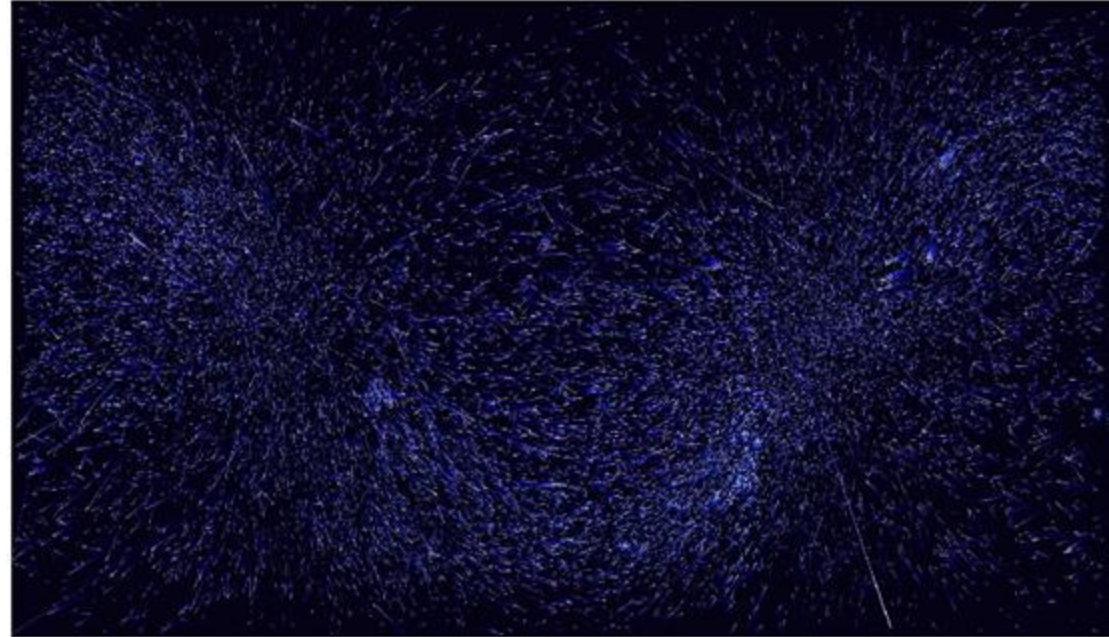
Variety

DIFFERENT FORMS OF DATA



Veracity: trustworthiness of data

- ◆ **Origin**
- ◆ **Authenticity**
- ◆ **Trustworthiness**
- ◆ **Completeness**
- ◆ **Integrity**



Types of Analytics (Plain English)

Descriptive — What happened?

Predictive — What is likely next?

Prescriptive — What should we do?

Different management problems are best solved by different analytical approaches . . .

THREE APPROACHES TO ANALYTICS			
Approach	DESCRIPTIVE Business Intelligence <i>What happened ?</i>	PREDICTIVE Prediction Engines <i>What will happen ?</i>	PRESCRIPTIVE Decision Automation <i>What should I do now ?</i>
Role of Machine	Helps me understand	Supports my decision	Tells me what to do
Size of Value-Creation Opportunity			
Examples	• Strategic Planning • Initial Product Pricing • Scenario Planning • Investor Reporting	• Demand Planning • Discount / Promotion mgmt. • CRM Segmentation • Maintenance	• Inventory Optimization • Price Optimization • Markdown optimization • Risk Optimization
Rationale	• Little data when compared with problem • High uncertainty • Simplified manual approach	• Quick-win opportunities • Relatively frequent decisions & observations • Semi-automation	• Larger Size of improvement prize • High frequency of decisions • Full automation

How do we solve business
problems with Data?



The Workflow: Decision → Metrics → Data → Analysis → Decision → Impact



Start with a decision, not with data



Choose a North Star KPI and supporting metrics



Define data needs (grain, owners, risks)



Analyse to inform an action; then measure impact & learn

North Star & KPI Tree



North Star: the primary definition of success



Supporting KPIs: levers & quality checks

Acquisition, activation, engagement, retention, monetisation

Operational quality (SLA, defect rates), customer experience



Guardrails protect margin, risk, and customer trust

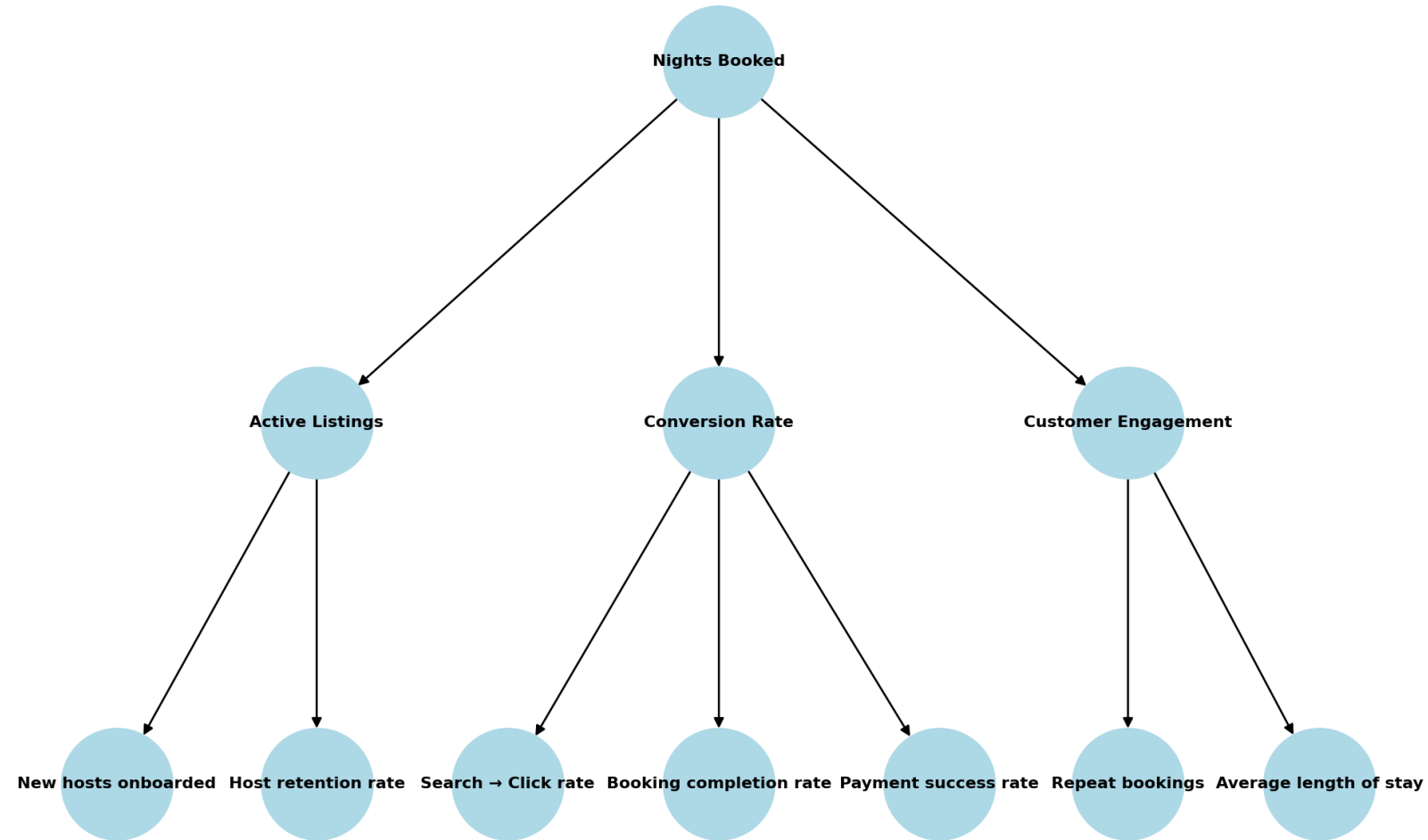
Example: AirBnB

- What is the North Star Metric for AirBnB?
- A **North Star Metric** is the single most important measure of long-term success — the one number that best captures the core value you deliver to customers.
- **Examples:**
 - Airbnb
 - Spotify:
 - Whatsapp:
 - Amazon:

North Star → KPI Trees

- A **KPI Tree** breaks down the North Star into **drivers and sub-metrics**. It shows how lower-level KPIs ladder up to the main goal.

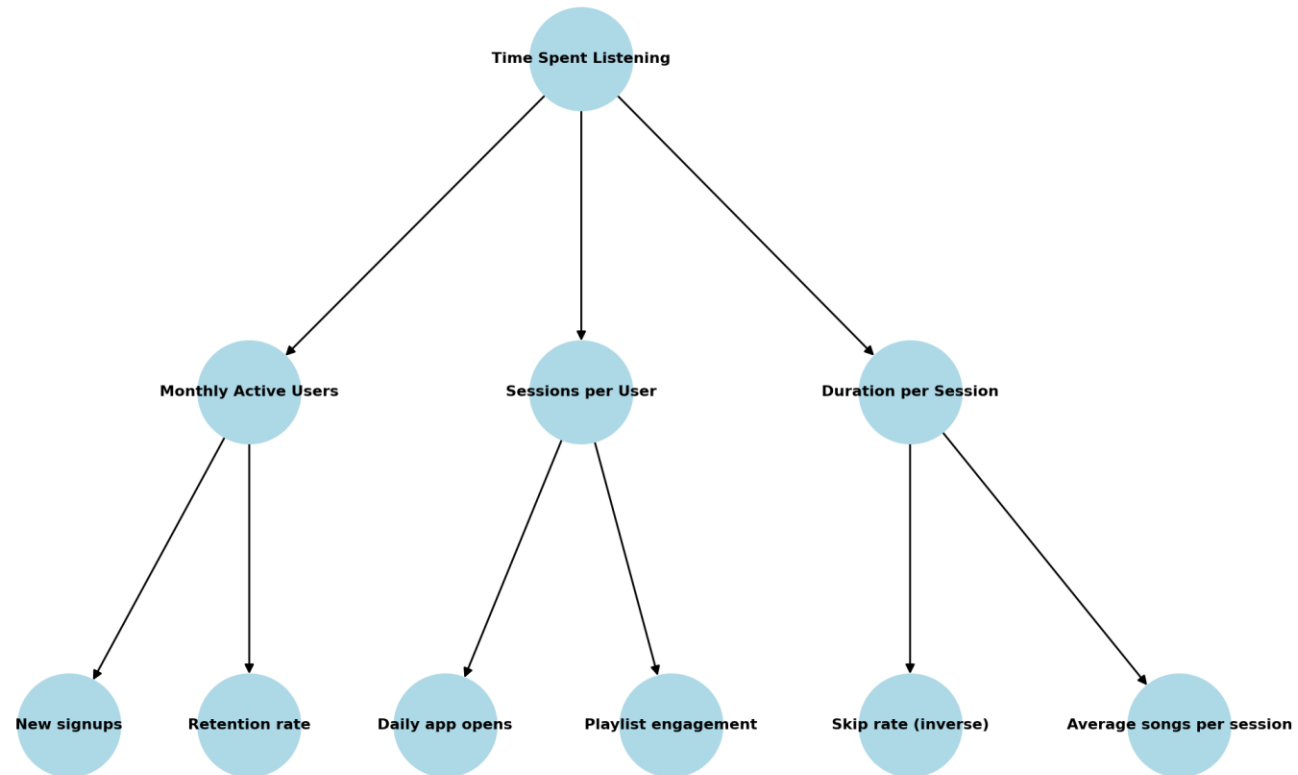
AirBnB KPI Tree



Spotify

- What is Spotify's North Star?
- Draw the KPI Tree for Spotify

Spotify KPI Tree



What Makes a Good Metric

Clear definition
(numerator/denominator, time
window)

Controllable by the team; aligned to
actions

Comparable over time/cohorts; stable
definitions

Trustworthy (hard to game; low
leakage/bias)

Data Basics for Managers

- Rows & columns; the concept of data 'grain'
- Keys & joins; sources & stewards/owners
- Common risks: missingness, sampling bias, data leakage

Definition of Grain: "What does one row in this dataset represent?"

Mini Data Dictionary (Template)

Name	Definition	Grain (row unit)	Source/Owner	Risks
active_user_28d	User with ≥ 1 session in last 28 days	user-day	Product Analytics	Bot traffic; inconsistent IDs
on_time_delivery_rate	% of orders delivered by promised time	order	Ops/Logistics	Clock sync; promise window drift
first_month_churn	% of new users inactive in first 30 days	user	CRM/Data	Definition drift; cohort mix

Visuals That Tell the Truth

- Match chart to question; avoid 3D and chartjunk
- Label clearly; show distributions, not just averages
- Beware dual axes unless necessary and clearly explained

GenAI as an Assistant (Not an Oracle)

- Good for: brainstorming KPI trees, metric definitions, checklists
- Not good for: ground-truth data, causality, sensitive/PII content
- Always disclose & verify GenAI-assisted outputs

Live Prompt (Demo)

- You are a business analyst. Create a KPI tree to reduce first-month churn for a subscription app.
- Include 1 North Star, 5 supporting KPIs with precise definitions (numerator/denominator, time window), 2 or 3 guardrails.
- Suggest likely data sources and flag risks (leakage/bias).
- Return the output as a table for an executive audience.

Exercise

- Pantaloons store is facing declining sales. You have been asked to solve the problem using data.
- Do the following:
 - Pick your decision
 - Draft your KPI tree: 1 North Star, 3–5 supporting KPIs, 2 guardrails
 - Write precise definitions (num/den, window) for each
 - Suggest likely data sources and flag risks (leakage/bias).
 - Return the output as a table for an executive audience.



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THANK YOU!